

Target clusters/
cluster centers

DP-sanitized
source prototypes

K Clusters

K Prototypes

$\{\nu_c\}_{c=1}^C \in \mathbb{R}^{d_t}$

$\{\mu_k^s\}_{k=1}^K \in \mathbb{R}^{d_s}$

Direct cross-domain distances are unreliable
Use only within-domain geometry

A 5x5 grid of squares. The top and left edges are labeled with indices: 1, 2, ..., c, C. The square at the intersection of row c and column c is highlighted with a red dashed border.

Diagram illustrating the mapping from K Prototypes to a set of vectors $\{\mu_k^s\}_{k=1}^K \in \mathbb{R}^{d_s}$.

From C_T :
take row c and sort
 c
Sort
(ascending)
 $\mathbf{s}_c^t = \text{sort}(C_T[c, :])$
 $\in \mathbb{R}^C$
 c

Row sorting removes
neighbour-order
ambiguity.

From C_S :
take row k and sort
 k 
Sort
(ascending)
 $\mathbf{s}_k^s = \text{sort}(C_S[k, :])$
 $\in \mathbb{R}^C$
 k 

Diagram illustrating the construction of the Relational cost matrix C_r .

Inputs: $\mathbf{s}_c^t$ (orange vector) and $\mathbf{s}_k^s$ (blue vector).

Output: Relational cost matrix C_r .

Formula for $C_r(c, k)$:

$$C_r(c, k) = \frac{\|\mathbf{s}_c^t - \mathbf{s}_k^s\|_2^2}{\max_{c', k'} \|\mathbf{s}_{c'}^t - \mathbf{s}_{k'}^s\|_2^2 + \epsilon}$$

Figure 1 shows a grid representing a cost matrix. The grid has rows indexed 1, 2, ..., c , C and columns indexed 1, 2, ..., k , K . A color bar on the right indicates 'High cost' (light green) and 'Low cost' (darker green). Several cells are highlighted with red borders, indicating high cost locations.

Partial Wasserstein plan

$$T^* = \arg \min_{T \in \Gamma_m(a,b)} \langle T, C_r \rangle_F$$

$$a = \mathbf{1}_C / C, \quad b = \mathbf{1}_K / K$$

Soft transport matrix T^*

1 2 ... k K

1 2 ... c C

1 0

<1 allows high confidence partial matching robust to DP noise/ambiguous clusters

A) Cluster-to-class bijection/
pseudo-labeling

Linear assignment on
row-normalized plan

$$\sigma = \arg \max_{\text{bijection}} \sum_{c=1}^C \bar{T}_{c, \sigma(c)}$$

Target clusters (C) Source classes (K)

1 → 1
2 → 2
C → K

pseudo-labels: $\hat{y}_j = \sigma(a_j)$

B) Transport conditions

For sample j in cluster $c = a_j$, use OT weights from $\bar{T}[c, :]$ to mix source prototypes.

$$\text{cond}_j^{\text{OT}} = \bar{T}[c, :] [\tilde{\mu}_{\sigma^{-1}(1)}^s, \dots, \tilde{\mu}_{\sigma^{-1}(K)}^s]^\top$$

0.27 0.54 0.20

$\text{cond}_j^{\text{OT}} \in \mathbb{R}^{d_s}$